Zirui Cheng

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Education

Princeton University, *Princeton, NJ* 2026 – Present
Ph.D. in Computer Science
Advisor: Peter Henderson

University of Illinois Urbana-Champaign, *Urbana, IL* 2024 – 2026
M.S. in Computer Science
GPA: 4.00/4.00

Tsinghua University, *Beijing, China* 2019 – 2024
B.Eng. in Computer Science and Technology
B.Econ. in Economics and Finance
GPA: 3.86/4.00

Research Interests

My research interests lie in ***machine learning***, ***natural language processing,*** and ***economics and computation***. I study language models as strategic agents. My research develops theoretical foundations and algorithmic methods for understanding and improving their agentic behavior. Through my research, I aim to address challenges related to their societal impacts (e.g., in law, economics, and politics) and alignment problems.

Publications

* *indicates equal contribution.*

[8] **Using Cognitive Models to Improve Language Model Simulation of Human Persuasion Games**
Zirui Cheng*, Zeyu Shen*, Thomas L. Griffiths, Peter Henderson.
Preprint, 2026.

[7] **Characterizing Rhetorical Misalignment in Decision-Making with Language Models**
Zirui Cheng, Joey Chan, Simo Du, Chenhao Tan, Yue Guo, Hao Peng.
Preprint, 2026.

[6] **Towards Strategic Persuasion with Language Models**
Zirui Cheng, Jiaxuan You.
ICLR 2026 – *Fourteenth International Conference on Learning Representations*.

[5] **ResearchTown: Simulator of Human Research Community**
Haofei Yu*, Zhaochen Hong*, **Zirui Cheng***, Kunlun Zhu*, Keyang Xuan, Jinwei Yao, Tao Feng, Jiaxuan You.
ICML 2025 – *Forty-second International Conference on Machine Learning*.

[4] **How Far Are We From AGI: Are LLMs All We Need?**
Tao Feng*, Chuanyang Jin*, Jingyu Liu*, Kunlun Zhu*, Haoqin Tu, **Zirui Cheng**, Guanyu Lin, Jiaxuan You.
TMLR 2024 – *Transactions of Machine Learning Research (10/2024)*.

[3] **Wikibench: Community-Driven Data Curation for AI Evaluation on Wikipedia**
Tzu-Sheng Kuo, Aaron Halfaker, **Zirui Cheng**, Jiwoo Kim, Meng-Hsin Wu, Sherry Tongshuang Wu, Kenneth Holstein, Haiyi Zhu.
CHI 2024 – *2024 ACM CHI Conference on Human Factors in Computing Systems*.

[2] **TreeQuestion: Assessing Conceptual Learning Outcomes with LLM-Generated Multiple-Choice Questions**
Zirui Cheng, Jingfei Xu, Haojian Jin.
CSCW 2024 – *2024 ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing*.

[1] **Modeling the Trade-off of Privacy Preservation and Activity Recognition on Low-Resolution Images**
Yuntao Wang*, **Zirui Cheng***, Xin Yi, Yan Kong, Xueyang Wang, Xuhai Xu, Yukang Yan, Chun Yu, Shwetak Patel, Yuanchun Shi.
CHI 2023 – *2023 ACM CHI Conference on Human Factors in Computing Systems*.

Research Experience

Princeton University, *Princeton, NJ* — 2025 Summer – Now
Research Assistant, Department of Computer Science
Advisor: Peter Henderson

University of Illinois Urbana-Champaign, *Urbana, IL* — 2025 Fall – 2026 Spring
Research Assistant, Siebel School of Computing and Data Science
Advisor: Hao Peng

University of Illinois Urbana-Champaign, *Urbana, IL* — 2024 Fall – 2025 Spring
Research Assistant, Siebel School of Computing and Data Science
Advisor: Jiaxuan You

Carnegie Mellon University, *Pittsburgh, PA* — 2023 Summer
Visiting Student, Human-Computer Interaction Institute,
School of Computer Science
Supported by Tsinghua University – Carnegie Mellon University Summer Program
Advisors: Kenneth Holstein, Haiyi Zhu

Tsinghua University, *Beijing, China* — 2022 Summer
Research Assistant, Department of Computer Science and Technology
Advisors: Yuntao Wang, Yuanchun Shi

Teaching Experience

CS222: Software Design Lab — 2025 Spring – 2026 Spring
Teaching Assistant
University of Illinois Urbana-Champaign

CS598: Deep Learning with Graphs — 2024 Fall
Teaching Assistant
University of Illinois Urbana-Champaign

Service

Conference Reviewer: ICLR 2027, NeurIPS 2026, ICML 2026, ICLR 2026, ACL 2026, UIST 2024.

Skills

Programming: Python, C/C++, JavaScript/CSS/HTML, Java, Verilog.

Languages: Chinese (native), English (proficient), German (fluent).